

DYNAMIC INVESTMENT STRATEGIES: PORTFOLIO INSURANCE VERSUS EFFICIENT FRONTIER

SERGEI ESIPOV

*Centre Solutions
a member of the Zurich Financial Services Group
One Chase Manhattan Plaza
New York, NY 10005*

IGOR VAYSBURD

*Martingale Technologies Inc
One Wall Street Court, Suite 300
New York, NY 10005*

ABSTRACT

Selected dynamic investment strategies are analyzed within a unifying theoretical framework. We suggest a Kolmogorov-type partial differential equation for a profit and loss (P&L) distribution of strategies contingent on the current value of the basic asset as well as on a *balance of a trading account* –“P&L-to-date”. This gives a possibility to study much wider class of strategies than is usually done in the literature and practical applications. Using our equation we build dynamic efficient frontier and demonstrate that an attempt to minimize variance for a given expected profit leads to a contrarian trading strategy, also known as the “St.Petersburg paradox”. Similar analysis is performed for the Black-Jones-Perold constant proportion portfolio insurance (CPPI). It is shown that both, dynamic efficient frontier and CPPI, belong to a special subclass of power option replicating strategies. Despite its small Sharpe ratio CPPI has an advantage of controlled downside, as its PL distribution is far from gaussian. Conversely, Sharpe-optimal dynamic efficient frontier has an uncontrollable downside. We show how to blend the advantages of these two strategies. A special part is devoted to discrete analysis and risk capital considerations for non-replicating strategies when P&L-to-date is not a single-valued function of the current asset value and path-dependency is essential. Finally, an example of analytically solvable cases for P&L distribution is given for hedging a European call with strategies, which depend on the balance of hedging account.

Keywords: Portfolio Insurance, Efficient Frontier, Profit and Loss Distribution, Basis Risk, Investment Strategies, Option Replication

I've often found that researchers who do not have practical experience with using financial models to make markets have much more faith in testing the predictability of models that is justified or practical. People working on Wall Street are more comfortable with the notion that a model provides a useful way of thinking.

E. Derman, "Reflections on Fischer" ¹

1. Introduction

Recently in ^{ii, iii} we suggested a novel analytical approach to studying contingent claims and hedging strategies. This approach allows one to study and, in some cases, analytically compute probability distributions of future profits or losses depending on the strategy followed by an investor or portfolio manager. In the course of derivation of the evolution equation for the probability distribution it was assumed that the prices of the tradable assets are lognormally distributed, and dynamic positions with respect to these assets are adjusted based on a strategy, which principles are defined in advance. Derivatives contingent on assets in the portfolio may influence the P&L distribution, since they may generate cash flows contingent on the assets at any intermediate/final times. The problem of pricing derivative contracts was set aside. It was assumed that the positions in these contracts are static, and they mature at some time horizon of interest, T , or earlier. Removal of these restrictions would imply that we have to take into account some additional risk factors, related to prices of derivatives. This problem will be addressed elsewhere.

Applicability of evolution equations derived in ^{ii, iii} (with the exception of Appendix B in ⁱⁱⁱ) is limited in one important aspect: dynamic control of the accumulated P&L of the trading account was not considered. Assuming that a given strategy generates P&L from F to $F + dF$ (measured in today's money) over the time interval $0 \leq t \leq T$ with probability P , we imply, that the fraction of random trajectories leading to F in terms of all possible trajectories is PdF . Here we ignore the following fact, far from being unimportant in financial practice, some of the above-mentioned trajectories may lead to a very small (negative) balance on the trading account, or its practical closure/liquidation at any intermediate time, t . For example, classical derivative hedging is anything but riskless given a borrowing constraint on capital, as it may lead to losses. The problem of studying the probability distribution of such losses appears, therefore, to be a quite natural one. The probability distribution that we are after is clearly different from what one can compute by using strategies insensitive to the balance of the trading account, i.e. by applying Eq(2.7) in ⁱⁱⁱ. In this paper we generalize Eq(2.7) in ⁱⁱⁱ by including the strategies dependent on the current accumulated balance, X . The resulting theory is rather flexible, it is critically dependent on the single assumption of Markovian nature of asset evolution, as log-normality can be replaced by the distribution of choice. Before we proceed with derivation, let us reiterate the basic assumptions of dynamical efficient markets.

Information flow surrounding dynamics of asset values is a process with diminishing time scales. It is regulated in such a way that the insiders have limited access to assets, while outsiders have limited access to the information. Majority of outsiders are, in fact, screened from the deterministic information as first explained in detail by Samuelson ^{iv}, and the market values/prices are considered as elements of a zero-sum game. In such a game the investors or practitioners acting on their behalf are facing the following problem. If the information of asset dynamics (known to insiders) suggests that the fundamental asset value will grow, the outsiders may be certain that the information chain between insiders and a given regular outsider has a sufficient capital support. By the time

the information becomes available to the outsider, the asset value is already up by a certain amount. This amount is believed to make the subsequent outsider investment volatile, leading to profit or loss, P&L. In markets with insufficient regulation, the situation may be different for well-capitalized investors in the middle of the information chain, this is not discussed here.

Investors using the public domain information are facing asset prices, which have been processed by the above-mentioned “filter”. These investors may safely assume that the reported market trend is already “priced in” the market valuesⁱ. Such markets are considered efficient, although the exact recipe of how a given trend is priced is unclear. The appropriate model of asset price dynamics is then the model without memory in the price process. Given that the parametrization of future asset dynamics is usually highly uncertain with the public domain information, when average trend and its variability is difficult to predict reliably, it is self-obvious that models containing asset drift and asset variance are probably as good as one can get. One arrives then at the concept of log-normal Brownian motion^v. More complex processes, including realistic histograms such as fractional Brownian motion are the next step. In most markets, fine features of these advanced models can be screened by uncertainty.

The fascinating property of Brownian motion is that it is simple to formulate and difficult to study in all details. All non-trivial strategies of investing/hedging/insuring with Brownian motion invoke P&L distributions, i.e. basis risk for all counter-parties. We indicate a general method, which is capable of handling risk of strategies dependent on asset prices *and* on accumulated P&L/available capital. Risk requires development of risk measures, and long-term analysis of consequences of applying such measures. In this paper we examine a variety of dynamic strategies applied to Brownian assets. We show that the dynamic problems may be quite different from the static ones. In the next Section we quickly reproduce the basic equations needed for analyzing P&L distributions of dynamic strategies. These equations are then applied to Markowitz-Sharpe mean-variance investment/asset allocation^{vi, vii, viii, ix} in Section 3. We show that the *dynamic efficient frontier* corresponds to the investment strategy known as the St.Petersburg paradox, when investor quickly unwinds the position as the profit increases, and increases the position as losses grow^x. This contrarian optimizer implies that the dynamic equivalent of CAPM (if it exists) not only has a problem associated with the arbitrary time horizon of investment, but may also cast shadows on the mean-variance approach for investors who are not prepared to use contrarian approach. Note that there exist outstanding examples of contrarian investment^{xi}.

Section 4 is devoted to strategies dependent on accumulated P&L, and relate exposure to the performance of the strategy. The *linear* (constant proportion) strategy of portfolio insurance (CPPI) by Black-Jones-Perold is in this category^{xii, xiii}. For earlier work on portfolio insurance see^{xiv, xv} for a review, see chapter 7 in Rubinstein's book^{xvi}. CPPI is characterized by high expected growth rates and even *higher* standard deviations of such rates. Therefore, the Sharpe's ratio of these strategies is low, and vanishes progressively with increasing their time horizon. Being useless from the mean-variance perspective the CPPI strategies have limited downside, and seem to be very attractive for investment purposes. The Dennis mechanical investment strategies seems to blend the both approached^{xvii}, and we give an example of how to do this by blending dynamic efficient frontier with CPPI. Portfolio insurance based on call replication is not discussed in this work as its expected return is smaller than for CPPI (see^{xiii}, p.405). Discrete formulation is presented Section 5, where we show that, in general, the P&L-dependent strategies are superior to strategies solely based on asset dynamics.

Before we proceed to quantitative details let us reiterate that in this work the feedback from the investment dynamics (back) to the asset dynamics is completely neglected. As a result we do not intend to model market bubbles/cycles. Moreover, the formation of market price, demand and supply considerations, and other economical factors are set aside for now. Participation in a given strategy may require substantial amount of introductory capital, and the results are dependent on its allocation.

Dynamics of asset price is most frequently described by a log-normal process. Readers with econometric and actuarial backgrounds may be uncomfortable with the log-normal model. Readers with practical experience in derivatives markets may feel that there is no simple connection between option theory and efficient frontier. We try to bridge this gap by admitting that whilst the analytical solution given below is certainly based on log-normality the procedure is generic. As always, to make a quantitative statement we assume that the uncertainty of the parametrization procedure is smaller than the optimization effect. Validity of this assumption can be verified by stress testing.

2. Brownian assets

Traditionally, a model Brownian asset at a market price S_t has a growth rate $mS_t t$ and variance $s^2 S_t^2$ reflecting that no other value scale but S_t is relevant. The latter assumption doesn't hold in alternative models. Establishing a "position" in the process

$$dS_t = mS_t dt + sS_t dW_t \quad (2.1)$$

based on the Wiener process W_t by holding j units of assets S_t leads to P&L up to time t , which we denote as X

$$X_t = \int_0^t e^{r(t-u)} j (dS_u - rS_u du). \quad (2.2)$$

Clearly, X_t and j may depend on many arguments (e.g. j may itself depend on X_t) and we do not write these dependencies explicitly. Here r is the risk-free rate (assumed to be free from fluctuations), and P&L is expressed in money values at time 0. Practitioners who make decisions depending on the asset values use (S_t, t) , while people who track their exposure prefer (X_t, t) . Discussions on the beneficial type of strategy can be extensive given the level of the problem. The process for X_t satisfies a stochastic differential equation

$$dX_t = rX_t dt + (m-r)j S_t dt + s j S_t dW_t \quad (2.3)$$

We consider strategies on the time interval $t \in [0, T]$. The system is fully described by the Green function: the joint probability density $Q_{\rightarrow}(X, S, t \rightarrow X', S', T)$ to arrive at S', X', T starting from S, X, t . This function is defined by normalization,

$$\int dS dX Q_{\rightarrow}(X, S, t \rightarrow X', S', T) = 1, \quad (2.4)$$

There exists the corresponding backward Green function which instead of initial is fixed by the final condition $Q_{\leftarrow}(X, S, T \leftarrow X', S', T) = \delta(X - X') \delta(S - S')$. Here $\delta(x)$ is the delta-function. For our purposes we need only a certain projection of this Green function,

$$P(F | X, S, t) = e^{r(T-t)} \int dS' Q_{\leftarrow}(X, S, t \leftarrow F e^{r(T-t)}, S', T). \quad (2.5)$$

This function is a conditional probability density¹ subject to normalization condition

$$\int dF P(F | X, S, t) = 1. \quad (2.6)$$

It satisfies almost the same backward Fokker-Planck equation as $Q_{\leftarrow}$ does. Namely, differentiating (2.5) with respect to time and using Eq(B.7) in ⁱⁱⁱ one finds

$$\frac{\partial P}{\partial t} + r \frac{\partial(FP)}{\partial F} + \hat{L}P = 0, \quad (2.7)$$

$$\hat{L} = [rX + (m-r)jS] \frac{\partial}{\partial X} + mS \frac{\partial}{\partial S} + \frac{1}{2} s^2 S^2 \left(j^2 \frac{\partial^2}{\partial X^2} + 2j \frac{\partial^2}{\partial S \partial X} + \frac{\partial^2}{\partial S^2} \right). \quad (2.8)$$

The solution of interest is specified by the final condition $P(F | X, S, T) = \delta(F - X)$ for investment strategies, or $P(F | X, S, T) = \delta[F - X + F_T(X, S)]$ for hedging long position in option, where $F_T(X, S)$ is the option payoff. Eq(2.7) is the basic equation of this work. When the strategy j is specified, the solution of Eq(2.7) gives the corresponding P&L distribution.

Along with Eq(2.7) for probability density of arriving at a certain profit or loss F , it is useful to derive the equations for the cumulants. The expected P&L $\bar{F}$, and variance V defined as,

$$\bar{F}(S, t) = \int dF F P(F | X, S, t), \quad V(S, t) = \int dF (F - \bar{F})^2 P(F | X, S, t)$$

satisfy the following backward equations

$$\frac{\partial \bar{F}}{\partial t} - r\bar{F} + \hat{L}\bar{F} = 0 \quad (2.9)$$

$$\frac{\partial V}{\partial t} - 2rV + \hat{L}V = -s^2 S^2 \left(j \frac{\partial \bar{F}}{\partial X} + \frac{\partial \bar{F}}{\partial S} \right)^2. \quad (2.10)$$

Note that the r.h.s. of (2.10) vanishes for

¹ Everywhere in this paper the term “distribution” implies “distribution density”.

$$j = -\frac{\partial \bar{F}}{\partial S} \bigg/ \frac{\partial \bar{F}}{\partial X}. \quad (2.11)$$

Substitution of this strategy into (2.9) leads to a nonlinear m -independent equation for the average P&L, $\bar{F}$. In general, unlike BS-equation, this nonlinear PDE describes expected P&L of a hedged contract with non-zero basis risk.

In the text we occasionally use terms hedging strategy, investment strategy, portfolio insurance strategy, while in all these cases we refer to the same mathematical object, j . This is explained by the fact that Eqs(2.7) or (A.5) are generic, while different terms refer to specific final or boundary conditions, or certain imposed constraints. Appendix A contains additional mathematical details on Eq(2.7) and discriminates between the so-called *replicating* and non-replicating strategies. Replicating strategies generate account balances X , which are path-independent for a given price of the underlying asset, S .

3. Strategies of the Dynamic Efficient Frontier

Efficient frontier and mean-variance optimization is a reliable and preferred tool for many practitioners who focus on Sharpe ratio along with other measures of risk. Strategies with high ratio and low correlation are always of interest for hedge and investment funds, proprietary trading desks, and many other groups of practitioners. Usually, these strategies are static in the sense that for a given time horizon (as suggested by liquidity) the long or short positions are kept unchanged.

The works by Markowitz and Sharpe allow practitioners to select an efficient portfolio based on returns and covariance matrix of individual assets, and the optimal portfolio has minimal standard deviation of profits and losses (P&L) for a given expected profit. If risk-less instruments are held, and manager can borrow money, the efficient frontier is a straight line on the plane expected total return (profit) versus its standard deviation.

In dynamics, even with a single asset, one can think that at every instant the “old” asset is (imaginary) sold and new one is bought. Let us put the transaction costs aside for now and form a *portfolio* of such instantaneous positions. Since each instantaneous position has its own return, variability and correlations with other instantaneous positions, one can apply mean-variance optimization to this portfolio. If the problem has a solution, it delivers a *dynamical efficient frontier*. Such frontier is superior to the classical static counterpart and has interesting properties, which we describe below.

Derivation of Dynamic Frontier

The probability of profit or loss F for an investment strategy $j(S, t)$, independent of the reserve, X , has a reduced form $P(F | X, S, t) = \tilde{P}(F - X | S, t)$. Its dynamics is governed by a partial-differential equation, which is immediately derived from (2.7) by substitution (we omit the tilde)

$$\frac{\partial P}{\partial t} + r \frac{\partial}{\partial F} (F + jS)P + mS \left(\frac{\partial P}{\partial S} - j \frac{\partial P}{\partial F} \right) + \frac{1}{2} s^2 S^2 \left(j^2 \frac{\partial^2 P}{\partial F^2} - 2j \frac{\partial^2 P}{\partial F \partial S} + \frac{\partial^2 P}{\partial S^2} \right) = 0, \quad (3.1)$$

and satisfies the terminal condition $P(F | S, T) = d(F)$. In the mean-variance framework our problem is to find such investment strategy, $j(S, t)$ that the expected NPV of P&L has a fixed value, $\bar{F} = F_0$, while the variance of NPV of P&L is minimal. The expected value $\bar{F}$ and variance, V , satisfy the equations (c.f. (2.8), (2.9))

$$\frac{\partial \bar{F}}{\partial t} + mS \frac{\partial \bar{F}}{\partial S} + \frac{1}{2} s^2 S^2 \frac{\partial^2 \bar{F}}{\partial S^2} - r\bar{F} = (r - m)jS, \quad (3.2)$$

$$\frac{\partial V}{\partial t} + mS \frac{\partial V}{\partial S} + \frac{1}{2} s^2 S^2 \frac{\partial^2 V}{\partial S^2} - 2rV = -s^2 S^2 \left(j + \frac{\partial \bar{F}}{\partial S} \right)^2. \quad (3.3)$$

The optimal strategy, denoted as $j_0(S, t)$, is

$$j_0(S, t) = \frac{aF_0}{sS_0(1 - e^{-a^2T})} \left[\frac{S}{S_0} e^{\frac{(m-3r)t}{2}} \right]^{-\frac{a+s}{s}} \quad (3.4)$$

$$a = \frac{m-r}{s}. \quad (3.5)$$

Here S_0 is the initial asset price. For the derivation of (3.4), (3.5) see Appendix B. We also found that this is a global optimal solution, which holds true in the full space of *path-dependent* strategies. Therefore, it is also the solution of the posed problem on the space $j(X, S, t)$. The minimal variance is simply

$$V_0 = \frac{F_0^2}{e^{a^2T} - 1}. \quad (3.6)$$

If the portfolio contains options, the above-given strategy is to be added to the Black-Scholes delta-hedge for the options.

Direct inspection shows that this strategy itself satisfies the equation for the Black-Scholes delta-hedge². Therefore the dynamic efficient frontier corresponds to a specific option replication as explained in Appendix A [strategy (3.4) leads to $H = 0$ in (A.3)]. The mapping at maturity between the accumulated P&L X and S is unique and given by the solution of the first of Eqs(A.4) with (3.4)-(3.5). The optimal strategy (3.4) replicates a *power* option with running P&L,

$$X(S, t) = \frac{F_0}{1 - e^{-a^2T}} \left[e^{rt} - e^{\frac{a(m-3r)t}{2s}} \left(\frac{S}{S_0} \right)^{-\frac{a}{s}} \right], \quad (3.7)$$

² This observation was first made by Alexander Lipton-Lifshitz when authors showed him formulae (3.4)-(3.5).

where integration constant was selected to ensure $X(S_0, 0) = 0$. To get the payoff of the power option one has to set $t = T$ in (3.7). Averaging (3.7) with the log-normal distribution for S and setting $t = T$ gives F_0 , as it should be. The profit with $j_0(S, t)$ is bounded, while the loss can be unlimited, as seen from (3.7) at vanishing S . For the numerical example with $m = 15\%$, $r = 5\%$, and, say, $s = 20\%$, Eq(3.7) gives the power option, $F(S_T) \propto S^m$, with power-law $m = -2.5$.

Properties of Dynamic Efficient Frontier

Let us discuss the optimal investment strategy $j_0(S, t)$.

1. An increase in asset price, $S > S_0$ is accompanied by a position *decrease* and vice versa (if we assume $m > r$). In other words, investors are advised to diminish the position as the asset price grows quickly, and to increase the position if the asset price plummets, as long as the decision to make the targeted profit is unchanged. While unwinding the position given a quick profit is commonplace, the position increase given a loss looks like a St. Petersburg paradox. To be precise, if the asset price diminishes in 2 times the position is increased in $2 \cdot 2^{(m-r)/s^2}$ times. The first serious study of this problem for the case $m = r = 0$ belongs to D. Bernoulli. Although expected return in Bernoulli problem is strictly zero, one can make the probability of loss arbitrarily small (in principle). The other name of this risky strategy is “martingale”.

2. Before addressing the paradox, one can confirm that the dynamic strategy contains less variance risk than the static one. Indeed, for a static position to buy and hold one asset, the expected P&L (in today's money) is $F_0 = S_0 e^{(m-r)T} - S_0$, and the variance is

$$V_S = S_0^2 e^{2(m-r)T} (e^{s^2 T} - 1). \quad (3.8)$$

Thus, the static variance is greater in

$$\frac{V_S}{V_0} = \frac{(e^{a^2 T} - 1)(e^{s^2 T} - 1)}{[1 - e^{-(m-r)T}]^2} \quad (3.9)$$

times than the dynamic one. At short times this ratio is close to 1, but at large times it can be exponentially large (and has a potential to overcome transaction costs).

3. The *static* Sharpe ratio of the static position is

$$\frac{F_0}{\sqrt{V_S}} = \frac{1 - e^{-(m-r)T}}{\sqrt{e^{s^2 T} - 1}}. \quad (3.10)$$

When the investment horizon changes, this ratio first increases, passes through a maximum, and then decreases. The optimal buy-and-hold horizon can be found numerically. In the case of dynamic strategy, the static ratio (for the same expected P&L) is

$$\frac{F_0}{\sqrt{V_0}} = \sqrt{e^{a^2 T} - 1} . \quad (3.11)$$

Unlike the static counterpart, it is always growing with time. For low growth rates and moderate volatility chosen in the numerical example above the effect of dynamic strategy is moderate, it takes a few years to benefit from it. However, the rate of re-balancing is small, and the dynamic strategy may make sense for pension funds with conservative risk profile and possibly for more aggressive funds in case of assets with enhanced returns. Application of static Sharpe ratio to the dynamic case is not immediately intuitive. It may be interpreted as an attempt to coarse-grain the description of investment process for senior fund management, as it replaces the details of the dynamic investment by a single static measure. It may also serve as a measure for the next level of contract aggregation in time. The instantaneous “microscopic” equivalent of Sharpe ratio can be identified as $\sqrt{T}(m - r) / s$. Note that the dependence of the optimal strategy (3.4) on T, m, r, s does not reduce to the dependence on this combination only, although that of (3.11) does.

4. The St. Petersburg paradox is fully resolved, for example, by capital considerations. This approach is formalized in the next section. (Another popular but less practical method is to introduce the utility function). It is important that one has to think about capital in this respect, since capital allocation is not a formalized issue, and it is surely different for different funds/regulation guidelines. To name one reason it is sufficient to mention that while a fund manager may plan to use dynamic optimization, senior management has to rely on preferred strategic yet approximate capital partition in view of uncertainty.

Given that for most dynamic strategies the return on capital and Sharpe ratio both depend on horizon it is impossible to make all horizons equal from investment viewpoint with a single price. This statement remains valid regardless of inter-relation between market price and asset fundamentals. If dynamic efficient frontier becomes popular this may lead to complete replacement of assets by a set of futures for every stock. Clearly, an asset to be kept in a portfolio for one year should be available at a discount as compared to asset positions to be terminated on the next business day. There exists a relation between these ideas and the liquidity preference theory^{xviii} which we do not discuss here.

4. Strategies which Depend on the Dynamically Available Capital

Whenever a practitioner considers hedging a portfolio, or looks for investment guidelines under given borrowing conditions, the available capital is always an important issue. Investment strategies, which explicitly depend on dynamically available capital, are considered in this section. The relation between these strategies, and efficient frontier (Section 3 above) can be found at the end of this Section.

We now return to Eq(2.7) and consider the Black-Jones-Perold CPPI strategy^{xii, xiii}. Translating their formula “exposure is a multiple of cushion”, $e = mc$, into our notations yields

$$j(X_t, S_t, t)S_t = m(X_t - X_R e^{rt}) . \quad (4.1)$$

Indeed, the cushion is understood here as the amount of risk capital that one can afford to lose. It is given by the difference between the dynamically available capital X_t , and the amount kept as a reserve, $X_R e^{rt}$. The latter, obviously, grows with the risk-free rate. The multiplier m is a constant number. Substitution of the portfolio insurance strategy (4.1) into Eq(2.7) yields

$$\frac{\partial P}{\partial t} + r \frac{\partial(FP)}{\partial F} + \left[rX + m(m-r)(X - X_R e^{rt}) \right] \frac{\partial P}{\partial X} + \frac{m^2}{2} s^2 (X - X_R e^{rt})^2 \frac{\partial^2 P}{\partial X^2} = 0, \quad (4.2)$$

where explicit S -dependence is omitted, since all other terms and the final condition, $P(F | X, t) = d(F - X)$ do not contain S . This equation is solved in Appendix C. P&L distribution for the excess profit $F - X_R e^{rt}$ is given by a log-normal distribution,

$$P(F | X, t) = \frac{1}{(F - X_R e^{rt}) \sqrt{2pm^2 s^2 (T-t)}} \exp \left[-\frac{A^2}{2m^2 s^2 (T-t)} \right], \quad (4.3)$$

$$A = \ln \frac{F - X_R e^{rt}}{X - X_R e^{rt}} - \left[m(m-r) - \frac{m^2 s^2}{2} \right] (T-t). \quad (4.4)$$

Thus, the expected profit over the entire time horizon is given by

$$\bar{F} = X_R + (X - X_R) e^{m(m-r)T}, \quad (4.5)$$

and P&L variance is given by

$$V_{BJ} = (X - X_R)^2 e^{2m(m-r)T} (e^{m^2 s^2 T} - 1). \quad (4.6)$$

The static Sharpe ratio of the BJ strategy for the matching static allocation $\bar{F} = F_0 = S_0 e^{(m-r)T} - S_0$ is

$$\frac{F_0}{\sqrt{V_{BJ}}} = \frac{S_0 e^{(m-r)T} - S_0}{[S_0 e^{(m-r)T} - S_0 - X_R] \sqrt{e^{m^2 s^2 T} - 1}}. \quad (4.7)$$

As stated in the introduction this ratio vanishes exponentially in time. At the same time the downside is clearly bounded by the reserve capital, see (4.3), (4.4). Obviously, the large variance comes from realizations with very high profits, which are inappropriately penalized in the Sharpe ratio. If one assumes that the investor is expecting a return on capital $\bar{m}$, or, in today's money, $\bar{m} - r$, formula (4.5) yields

$$m = \frac{\bar{m} - r}{m - r} . \quad (4.8)$$

Using expected return on capital $\bar{m} = 30\%$, market return $m = 15\%$, and risk-free rate $r = 5\%$ one finds $m = 2.5$, which is directly the opposite of what we found in Section 3. This is indeed the numerical value of the multiplier suggested by Black and Jones^{xii}. Given that the downside is protected, further optimization over m is usually based on maximizing the expected return. We first note that the expected return exponentially grows with m without bounds. Here the St. Petersburg paradox is inverted: the probability p to achieve return, which is comparable or exceeds the expected P&L, becomes progressively small, $\ln p \sim -ms\sqrt{T}$. In other words, investment strategies with large m are similar to lottery. Strategies with $m > 1$ may also require unlimited borrowing of capital³. The boundary case $m = 1$ is the trivial static allocation, $j = \text{const}$.

Another way to approach the constant proportion portfolio insurance is to solve Eq(2.15) and look for unique mapping between X and S . The solution of

$$\frac{\partial X}{\partial S} = \frac{m(X - X_R e^{rt})}{S} \quad (4.9)$$

reads $X(S, t) = S^m C(t) + X_R e^{rt}$, where $C(t)$ is an arbitrary function. This solution does satisfy the BS-equation provided

$$C(t) = C_0 \exp[-(m-1)(r + \frac{1}{2} m s^2)t], \quad (4.10)$$

where the integration constant $C_0 = (X_0 - X_R)S_0^{-m}$ is selected so that $X(S_0, 0) = X_0$. Thus, the CPPI strategy is also a power option replicator (c.f. section 3), with the positive power-law payoff $F_T(S) \propto S^m$, see^{xiii} for additional details. The capital X_R is separated into a completely independent reserve. Computing expectation of $X(S, T)$ over log-normal distribution of S and discounting to $t = 0$ one arrives at (4.5), as it should be. At this point the theory of dynamic efficient frontier and CPPI are merged into the same class of strategies, replicating power options. The remaining difference between the two is equivalent to taking long or short position in St. Petersburg game.

By using mapping to convert from X – to S – dependencies one can solve some optimization problems, as long as mapping exists. The blended variance-optimization problem in the full X, S space based on Eq(2.7) with a provision $X \geq X_R e^{rt}$ represents an example where mapping is unclear *a priori*. The problem is difficult because the coefficients in (2.9), (2.10) unlike in (3.2), (3.3) explicitly depend on $j(X, S, t)$, and numerical methods may be required. We now revisit risk and return on trees.

5. Discrete Strategies and Optimization

As various Value-at-Risk measures gain increasing popularity, it is frequently debated that practitioners need the ultimate set of Values-at-Risk, i.e. the full P&L distribution. In their turn, P&L distributions become *conditional* on the portfolio-wide dynamic hedging strategies. The main message of this section is the discrete method with which one can analyze future risk. We begin with a portfolio *contingent* on only one major index evolving on a binary tree, Fig 1. Let it be assumed that the index of value S can move (up or down) to only two new possible values, S_1, S_2 with *objective* probabilities p and $1-p$ for two possible future states of the world. Let the contingent portfolio with known market value (MV) have certain expected values and variances in these states, see Fig 1.

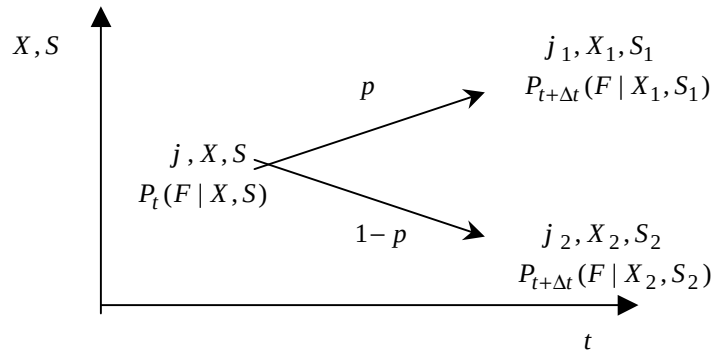


Fig.1. One time-step on the binary tree

If the above-specified odds are correct, the portfolio is at risk *today* even if its market value states the contrary. To demonstrate this explicitly, let us roll the expected value and its variance one step Δt backward. If the proper inverse “zero” used for discounting is denoted as $Z = \exp(-r\Delta t)$, one finds the net present value of expected future state

$$Z\bar{F} = p\bar{F}_1 + (1-p)\bar{F}_2, \quad (5.1)$$

and similar for the squared present value

$$Z^2\overline{F^2} = p\overline{F_1^2} + (1-p)\overline{F_2^2}. \quad (5.2)$$

Raising (5.1) to the second power and subtracting from (5.2) yields the variance, $V = \overline{F^2} - \bar{F}^2$,

$$Z^2V = pV_1 + (1-p)V_2 + p(1-p)(\bar{F}_1 - \bar{F}_2)^2. \quad (5.3)$$

³ These strategies are progressively vulnerable to market jumps (neglected in the log-normal model).

This is the variance in today's terms, which can be rolled on the tree. If the underlying index S is tradable, it is possible to control variance, at least partially. Namely, consider purchasing j shares at present and selling them in the future. For the “new” portfolio consisting of the “old” portfolio and these shares (5.1), (5.2) become dependent on j

$$Z\bar{F} = p\bar{F}_1 + (1-p)\bar{F}_2 + j_w [pS_1 + (1-p)S_2 - ZS], \quad (5.4)$$

$$Z^2V = pV_1 + (1-p)V_2 + p(1-p)(S_1 - S_2)^2 \left(j + \frac{\bar{F}_1 - \bar{F}_2}{S_1 - S_2} \right)^2. \quad (5.5)$$

Eq(5.4) shows that the expected value is shifted by expected profit or loss on the hedge, as it should be. Formula (5.5) shows how the variance of the net portfolio (with shares) can be controlled. Eqs(5.4), (5.5) are discrete counterparts of Eqs(2.9), (2.10). To cancel the variance one has to hold j shares,

$$j = -\frac{\bar{F}_1 - \bar{F}_2}{S_1 - S_2}. \quad (5.6)$$

We hope the reader will note the fact that the celebrated Δ – hedge naturally appears here for the sample portfolio not containing any options.

Let us now consider the efficient frontier for this single time step. Solving (5.4) for j and substituting into (5.5) results in

$$Z^2V = pV_1 + (1-p)V_2 + p(1-p) \left[\frac{Z\bar{F}(S_1 - S_2) - ZS(\bar{F}_1 - \bar{F}_2) + \bar{F}_1S_2 - \bar{F}_2S_1}{pS_1 + (1-p)S_2 - ZS} \right]^2. \quad (5.7)$$

The dependence of portfolio's standard deviation, $\sqrt{V}$ versus its expected value, $\bar{F}$ yields a parabolic curve, where one may select the preferred risk profile. Returning to the variance-minimizing Δ – hedge (5.6), and substituting it into (5.4) one finds

$$Z\bar{F} = ZS \frac{\bar{F}_1 - \bar{F}_2}{S_1 - S_2} + \frac{\bar{F}_2S_1 - \bar{F}_1S_2}{S_1 - S_2}. \quad (5.8)$$

The well-known property of this expression is that it does not depend on objective probability, p . Eq(5.5) yields for the minimal variance

$$Z^2V = pV_1 + (1-p)V_2. \quad (5.9)$$

This algebra is illustrated with simple numerical example in Appendix D. The power of the method becomes immediately clear with portfolios containing options, as the method does *not* change at all, and can accommodate any set of cash flows.

Multi-step dynamic hedging is performed by making use of recombining or non-recombining trees, with “many Figs 1” linked together. Each node is characterized by its

asset value, S , and full distribution of the portfolio P&L, $P_t(F | X, S)$, along with the corresponding mean, variance, and so on. The discrete analog of Eq(2.7) is

$$P_t(Z^{-1}F | Z^{-1}X, S) = pP_{t+\Delta t}[F | X - j(S_1 - ZS), S_1] + (1-p)P_{t+\Delta t}[F | X - j(S_2 - ZS), S_2]. \quad (5.9)$$

The numerical problem is supplied by the discrete final condition $P(F | X, S) = d_{FX}$, where d is the Kronecker symbol, and boundary condition, for example, $P(F | Z^{N-n}X_R, S) = 0$, similar to a knock-out option. Here N is the total number of backward time steps, and n is the current step.

It is straightforward to blend DEF and CPPI on trees by using any robust optimization technique for minimizing the variance of numerical solution of (5.9) over the set of j . As an example of numerical solution of Eq(5.9) we consider the following problem, related to the works by Föllmer and Leukert^{xix}, see also Karatzas and Cvitanic^{xx}. These authors consider a hedging strategy, which minimizes the variance and/or the probability given $X_T + F_T(S) < 0$ (in our notations) when hedging a contingent claim, $P(F | X, S, T) = d[F - X - F_T(S)]$, under a constraint $X \geq 0$. Here $F_T(S)$ is the payoff of the claim, and the initial capital is $X_0 > 0$. The solution, according to the authors of^{xix, xx} is to make use of the Black-Scholes hedge for a special combination of options including a knock-out option. There always exists a possibility that a non-replicating strategy on the space parametrized by (X, S, t) gives a better solution, and a special proof is required for a replicating solution that the corresponding minimum is the global one.

Below we focus on a version of blended strategy with a European call option at maturity, which displays the advantages of CPPI and dynamic efficient frontier. Suppose a European call is sold, and the writer sets up a hedging account with a given initial capital X_0 (not necessarily equal to the proceeds of the sale). This account is used for maintaining the hedge, and at no time this account should be negative with the exception of maturity date when the option pay-out may lead to a loss. Mathematically, in the simple case $r = 0$, this condition is expressed as

$$X_0 + \int_0^t j_u dS_u \geq j_t S_t, \text{ or } X_t \geq 0, \quad (5.10)$$

where $j_t \equiv j(X, S, t)$ is the hedge at time t . Condition (5.10) means that the initial capital plus profit/loss from hedging is always sufficient to afford the hedge. Condition (5.10) can always be satisfied for large enough initial capital. With large X_0 there is no theoretical risk in the transaction, and perfect BS-replication is optimal. Fig. 2 shows a contour map of discrete BS-hedge, j for this case at every node of a tree with only 3 time steps (which can also be studied analytically).

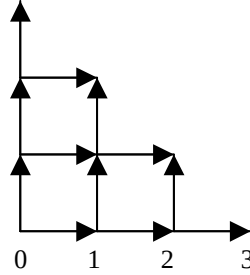


Fig.2. The 3 step binomial tree. Numerical parameters:

$$S_0 = 1, X_0 = 1.2, F_T = (S - 1)^+, p = 0.6, u = 1.2, T = 3.$$

u is the ratio of two successive asset prices if the price goes up.

We consider three types of strategies. The first is BS strategy, which satisfies the discrete analog of BS-equation for Δ -hedge:

$$j(1+u) = j_1 u + j_2, \quad (5.11)$$

where $j_{1,2}$ are indicated in Fig.1. The second is $j(S, t)$ strategy on the tree Fig.2, which does not necessarily replicate an option. The third is the $j(X, S, t)$ strategy, which in this case can be thought of as a strategy on a non-recombining tree. There is only one node on this tree at $S = 1, t = 2$ where the accumulated P&L X depends on whether this point is reached by moving up and then right, or by moving right and then up. At large enough $X_0 > 1.136$ there is no difference between the trees, and the optimal strategy is a BS-hedge.

Time step	S	X	j
0	1.00	0.57	0.57
1	1.20	0.30	0.79
	0.83	0.79	0.30
2	1.44	0.00	1.00
	1.00	0.55	0.55
	1.00	0.55	0.55
	0.69	1.00	0.00

Table.1 Asset price S , available capital, X , and optimal hedge j for $X_0 = 1.136$.

Note that all values of X are non-negative.

At smaller values of $0 \leq X_0 \leq 1.136$ the three strategies differ, in general. For example, at $X_0 = 0.6$, $\langle X \rangle = 0.45$ the BS strategy is

Time step	S	X	j
0	1.00	0.06	0.54
1	1.20	0.00	0.59
	0.83	0.11	0.48
2	1.44	0.00	0.59
	1.00	0.00	0.59
	1.00	0.00	0.59
	0.69	0.20	0.35

Table.2 Asset price S , available capital, X , and optimal hedge j for $X_0 = 0.6$, $\langle X \rangle = 0.45$. Note equal values of $X(t = 2, S = 1) = 0$ for both paths (shaded). This BS-strategy has no defaults, and residual standard deviation of P&L $s = 0.08670$.

Here $\langle X \rangle$ is computed after the option liability has been paid. The next table shows the optimal $j(S, t)$ strategy.

Time step	S	X	j
0	1.00	0.00	0.60
1	1.20	0.00	0.60
	0.83	0.18	0.39
2	1.44	0.00	0.60
	1.00	0.04	0.56
	1.00	0.00	0.56
	0.69	0.36	0.12

Table.3 Asset price S , available capital, X , and optimal hedge j for $X_0 = 0.6$, $\langle X \rangle = 0.45$. Note *non-equal* values of $X(t = 2, S = 1)$ for both paths, but still equal values of $j(S = 1, t = 2) = 0.564$ (shaded). This non-BS-strategy has no defaults, and residual standard deviation of P&L $s = 0.07787$.

The last table shows the optimal $j(X, S, t)$ strategy.

Time step	S	X	j
0	1.00	0.00	0.600
1	1.20	0.00	0.60
	0.83	0.18	0.39
2	1.44	0.00	0.600

	1.00	0.06	0.54
	1.00	0.00	0.57
	0.70	0.35	0.14

Table.3 Asset price S , available capital, X , and optimal hedge j for $X_0 = 0.4$.
 Note non-equal values of $X(t = 2, S = 1)$ and *non-equal* values of $j(S = 1, t = 2)$ for both paths. This non-BS-strategy has no defaults, and residual standard deviation of P&L $s = 0.07783$.

The behavior at constraining X_0 can be complex, with many competing local optimal solutions. Frequently, small changes in X_0 may lead to significant changes in (local) optimal arrangement (not shown). If “annealing” is terminated and one now gradually increases X_0 the system finally “emerges” from the alternative solutions and returns to the no-risk case. A hysteresis is not necessarily repeated in cycles even if the ΔX_0 steps are maintained unchanged, but a small offset of the initial point is made. For different sets of steps the system may display completely different configurations. At lower values of X_0 the optimal hedge may look disordered (or what is called “glassy” in real-world annealing). Thus, even with Eq(5.9) the numerical problem of finding the optimal strategy can be quite difficult, not mentioning Monte-Carlo simulations. Many of the local minima can be improved by invoking X – dependence, however, to begin with, the numerical optimizing routine can be easily trapped in a sub-optimal state. Complexity of the landscapes suggests that their finer details are progressively model-sensitive. Nevertheless, the statistical study of glass-like landscapes and characterization of their density of states is an interesting problem.

Thus, we have shown numerically how to optimize the variance for a given mean provided that there are imbedded options and the downside is protected at any time.

6. An Analytically Solvable Example

In this section the solution of the basic equation Eq(2.7) is presented for the type of problems where position in the underlying asset is maintained static until the trading account runs out of capital. We consider three cases.

Case (i). Suppose that a European call option is sold and this position is “hedged” by holding a constant number of underlying assets, denoted as Δ . The trading account, where the hedge position is monitored, initially has a certain amount of capital, $X_0 > 0$. If this capital is depleted, i.e. at some time $t < T$ the balance of the trading account becomes zero, the hedging position is liquidated, and an offsetting option is purchased (at then current Black-Scholes price), so that the seller can “walk away” with an achieved profit or loss, F .

We also consider the situation when the option is bought, and the position is “hedged” with the static number of shares, Δ . As before, if we run out of capital, the hedging account is liquidated and the position in the option is (ii) liquidated as well, (iii) kept till maturity. Arithmetic diffusion is assumed below, with diffusivity $s^2/2$.

Introducing notations

$$\begin{aligned}
r(x, t) &= \frac{1}{\sqrt{2ps^2t}} e^{-\frac{x^2}{2s^2t}}, \\
\tilde{r}_{X,\Delta}(S, t \rightarrow S', t') &= r(S' - S, t' - t) - r(S' - S + \frac{2X}{\Delta}, t' - t), \\
\Omega_\Delta(t' | X, S, t) &= \frac{X}{t' - t} r\left(\frac{X}{\Delta}, t' - t\right), \\
F_{BS}(S, t) &= f(S, t),
\end{aligned} \tag{6.1}$$

where F_{BS} is the Black-Scholes call option price with the strike K , maturity T , and the constant risk-free rate r , and

$$\int_t^\infty dt' \Omega_\Delta(t' | X, S, t) = 1, \tag{6.2}$$

one can show that the solution of Eq(2.7), i.e. the running P&L distribution, is given by the following formula

$$\begin{aligned}
P_\Delta(F | X, S, t) &= q[X + \Delta(K - S) - F] \left[\frac{q(F)}{\Delta} \tilde{r}_{X,\Delta}(S, t \rightarrow \frac{F - X}{\Delta} + S, T) + \right. \\
&\quad \left. + \frac{1}{1 - \Delta} \tilde{r}_{X,\Delta}(S, t \rightarrow \frac{F - X - K}{\Delta} + S, T) \right] + \\
&+ q[F + f(S - \frac{X}{\Delta}, t)] q(-F) \left| \frac{\partial F_{BS}(S - X/\Delta, t^*)}{\partial t} \right|^{-1} \Omega_\Delta(t^* | X, S, t),
\end{aligned} \tag{6.3}$$

where the root $t^*(F)$ is the solution of the equation

$$F + f(S - \frac{X}{\Delta}, t^*) = 0, \quad 0 \leq t^*(F) \leq T. \tag{6.4}$$

It is assumed in (6.3) that $K > S - X/\Delta$, and $0 < \Delta < 1$. In the other two cases one finds

$$\begin{aligned}
P_\Delta(F | X, S, t) &= q[F - X - \Delta(K - S)] \left[\frac{1}{\Delta} \tilde{r}(S, t \rightarrow \frac{F - X}{\Delta} + S, T) + \right. \\
&\quad \left. + \frac{q(S - K - F - X/\Delta)}{1 + \Delta} \tilde{r}(S, t \rightarrow \frac{F + K - X}{1 + \Delta} + S, T) \right] \\
&+ \int_t^T dt' \Omega_\Delta(t' | X, S, t) P_0(F | S - \frac{X}{\Delta}, t'),
\end{aligned} \tag{6.5}$$

where we denoted

$$P_0(F | S, t) = d[F - f(S, t)], \quad (6.6)$$

in case (ii), and

$$P_0(F | S, t) = q(F) r(K + F - S, T - t) + d(F) \int_{-\infty}^K dS' r(S' - S, T - t) \quad (6.7)$$

in case (iii). It is assumed in (6.5) that $K > S - X / \Delta$, and $-1 < \Delta < 0$. The properties of these P&L distribution densities will be discussed elsewhere.

7. Conclusion

We have presented a general theory, based on Eq(2.7), which allows one to study dynamic efficient frontier (DEF), constant-proportion portfolio insurance (CPPI), and various strategies of incomplete markets in a unifying framework. It was found that both DEF and CPPI admit perfect replication and belong to the same class of power options. The objective of the analysis is to derive the today's estimate of the future P&L distribution and use it for resolving all issues associated with single-number risk measures. It was found that for a specified time horizon the P&L distribution of DEF has a risky "tail" with unlimited downside. The attractive CPPI P&L distributions with high profits and limited downside are inadequately penalized in terms of their small Sharpe ratio. However, their high profits may come from lottery-like settings where investor effectively takes a short position in St. Petersburg game. It is somewhat instructive that this game surfaces at the turn of the XX century in the problem of dynamic optimization in stochastic calculus. Novel strategies which blend advantages of DEF and CPPI are obtained by numerical solution of Eq(2.7). Although the long-term dynamics of capital in an investment fund may become normally distributed, each individual strategy and risk management can benefit from protected downside. We think that the techniques presented above will lead to further development of innovative risk-controlled investment strategies.

7. Acknowledgments

SE is thankful to Richard Timbrell and Dajiang Guo for help with the literature and useful conversations, to Jay Blumenstein for providing the reference ^{xix}, and especially to Alexander Adamchuk for discussions and help with a number of formulae in Sections 3, 5 and Appendixes A, B, including correction of few errors.

Opinions expressed in this article are the authors' and do not necessarily correspond to the views of Martingale Technologies Inc. or Centre Solutions and Zurich Financial Services Group or any of their subsidiaries. All errors in this text are our responsibility.

Appendix A. Miscellaneous Mathematical Properties of Eq(2.7)

The differential operator $\hat{L}$ is parabolic with zero discriminant. Local diagonalization is achieved by transformation

$$X' = X - \tilde{F}(S, X, t), \quad \frac{\partial \tilde{F}}{\partial S} + j \frac{\partial \tilde{F}}{\partial X} = j. \quad (\text{A.1})$$

The diagonalized operator reads

$$\hat{L}' = mS \frac{\partial}{\partial S} + \frac{s^2 S^2}{2} \frac{\partial^2}{\partial S^2} - H \frac{\partial P}{\partial X'}, \quad (\text{A.2})$$

where

$$H = \frac{\partial \tilde{F}}{\partial t} + \frac{\partial \tilde{F}}{\partial S} \left(rS - \frac{rX}{j} \right) + \frac{s^2 S^2}{2} \left(j^2 \frac{\partial^2 \tilde{F}}{\partial X^2} + 2j \frac{\partial^2 \tilde{F}}{\partial X \partial S} + \frac{\partial^2 \tilde{F}}{\partial S^2} \right). \quad (\text{A.3})$$

In (A.3) one has to replace X by X' by inverting (A.1). General solution of the diagonalization constraint in (A.1) with characteristics

$$\frac{dX}{dS} = j, \quad \frac{d\tilde{F}}{dX} = 1 \quad (\text{A.4})$$

is fixed by using boundary and/or initial conditions. Eqs(2.7), (2.9), (2.10) yield after transformation

$$\frac{\partial P}{\partial t} + r \frac{\partial(FP)}{\partial F} + \hat{L}' P = 0, \quad (\text{A.5})$$

$$\frac{\partial \bar{F}}{\partial t} - r\bar{F} + \hat{L}' \bar{F} = 0, \quad (\text{A.6})$$

$$\frac{\partial V}{\partial t} - 2rV + \hat{L}' V = -s^2 S^2 \left(\frac{\partial \bar{F}}{\partial S} \right)^2. \quad (\text{A.7})$$

If Eqs(A.4) are solved, and function H is found to be independent on S , the X -dependence and S -dependence factorize in the Green function of the operator $\hat{L}'$. Then (A.5) can be analytically integrated. If, in addition, $H = 0$ then the strategy leads to *unique* $X \leftrightarrow S$ mapping. For the strategies independent on X the condition $H = 0$ yields the BS-equation for $\tilde{F}$ or BS-equation of the Δ -hedge for j .

Note that for a general choice of $j(X, S, t)$ the operator $\hat{L}$ does not conserve the operand. The question of variance minimization can be studied on the basis of

X – dependent Eqs(2.7)-(2.8). Comparing Eqs(2.10) and (3.3) one finds a close analogy between (2.11) and the BS strategy $j(S, t) = \partial \bar{F} / \partial S$. These strategies minimize the right-hand-sides of (2.10) and (3.3), respectively. While the BS-strategy clearly minimizes variance in the (S, t) -space, since the transport operator in (3.3) has j – independent coefficients, the benefits of (2.11) are unclear, *a priori*.

Consider a scenario where terminal condition at maturity does not contain residual variance, $V(X, S, T) = 0$, and strategy (2.11) is selected, which sets the r.h.s. of Eq(2.10) to zero. Integrating Eq(2.10) over (X, S) one finds, that the sufficient condition for keeping zero variance is

$$(m - r)Sj_X = s^2(S^2jj_{XX} + S^2j_X^2 + S^2j_{SX} + 2Sj_X), \quad (\text{A.8})$$

provided that the variance is not supplied from the boundary. Integrating (A.8) with respect to X one finds,

$$jj_X + j_S = \frac{aj}{S}, \quad a = \frac{m - r - 2s^2}{s^2}, \quad (\text{A.9})$$

where the integration constant is set to zero at a distant boundary. This first-order PDE has the following characteristics

$$\frac{dX}{dS} = j, \quad \frac{dX}{dj} = \frac{jS}{a}, \quad (\text{A.10})$$

with one of them being identical to (A.1). However, the other characteristic is novel. As a result, the significance of the hedge (2.11) is unclear at this stage, as in general X – dependent case it generates variance. The only interesting property of (2.11), as is pointed out in the main text, is that substitution of this strategy into (2.9) leads to a nonlinear m -independent equation for the average P&L, $\bar{F}$.

Note that the second-order WKB-approximation of Eq(2.7) is equivalent to solving (2.9), (2.10).

Appendix B. Detailed Derivation of Dynamic Efficient Frontier

Consider the system (3.2)-(3.3) from the main text, subject to final conditions $\bar{F}(S, T) = V(S, T) = 0$ and $\bar{F}(S_0, 0) = F_0$. Let it be required to find such a strategy $j(S, t)$ on the space (S, t) , which minimizes the variance $V(S_0, 0)$. This problem is

solved below by the variational procedure ⁴. Using log-price, $y = \ln(S/S_0)$, and denoting exposure as $f = Sj$ one can re-write (3.2)-(3.3) as follows

$$\frac{\partial \bar{F}}{\partial t} + h \frac{\partial \bar{F}}{\partial y} + \frac{1}{2} s^2 \frac{\partial^2 \bar{F}}{\partial y^2} - r \bar{F} = (r - m) f, \quad (\text{B.1})$$

$$\frac{\partial V}{\partial t} + h \frac{\partial V}{\partial y} + \frac{1}{2} s^2 \frac{\partial^2 V}{\partial y^2} - 2rV = -s^2 \left(f + \frac{\partial \bar{F}}{\partial y} \right)^2, \quad (\text{B.2})$$

where $h = m - s^2/2$. Let $r_{ww'}^{(n)}$ be the Green function of the backward diffusion equation with drift and decay,

$$\frac{\partial r^{(n)}}{\partial t} + h \frac{\partial r^{(n)}}{\partial y} + \frac{1}{2} s^2 \frac{\partial^2 r^{(n)}}{\partial y^2} - nr r^{(n)} = \delta(y - y') \delta(t - t'), \quad (\text{B.3})$$

and let w denote the pair (y, t) . This function has the explicit form

$$r_{ww'}^{(n)} = \frac{1}{\sqrt{2ps^2(t'-t)}} \exp \left\{ -\frac{[(y-y')-h(t'-t)]^2}{2s^2(t'-t)} - nr(t'-t) \right\}, \quad (\text{B.4})$$

and its Fourier transform in y is

$$\tilde{r}^{(n)} = \exp \left[(t'-t) \left(-\frac{s^2 k^2}{2} + i h k - nr \right) + i k y' \right]. \quad (\text{B.5})$$

Solving Eq(B.1) yields

$$F_w = (m - r) r_{ww'}^{(1)} f_{w'}. \quad (\text{B.6})$$

Here repeating indices indicate integration. (B.6) allows one to specify the constraint on the expected profit

$$(m - r) r_{w_0 w'}^{(1)} f_{w'} = F_0. \quad (\text{B.7})$$

⁴ Solution of such problems by using locally optimal strategies satisfying the Hamilton-Jacobi-Bellman (HJB) equation may lead to sub-optimal results as the global mean value is not preserved. This is not the case only for simple variance minimization as discussed by Hipp and Taksar who attempted to solve our equation for variance by the HJ method

Note that this constraint is non-local, and that is why we use the global variational method instead of the local (Hamilton-Jacobi-like) equation. The variance is then given by

$$V_w = s^2 r_{ww}^{(2)} [f_w + \partial_{y'}(m-r)r_{w'w''}^{(1)} f_{w''}]^2, \quad (\text{B.8})$$

and the relevant Lagrangian form is

$$L_{w_0} = V_{w_0} - l(\bar{F}_{w_0} - F_0), \quad (\text{B.9})$$

where l is the Lagrange multiplier. The functional derivative of L_{w_0} wrt f_n from (B.6)-(B.9) reads

$$2s^2 r_{w_0w}^{(2)} [f_w + \partial_{y'}(m-r)r_{w'w''}^{(1)} f_{w''}] [d(w'-n) + \partial_{y'}(m-r)r_{w'w''}^{(1)} d(w''-n)] - l(m-r)r_{w_0w}^{(1)} d(w'-n) = 0. \quad (\text{B.10})$$

This linear integral equation has to be solved along with (B.5). As it frequently happens, there is more than one way to solve a solvable problem, and we make use of the one with enjoyable algebra. Close inspection of (B.8) shows that the function,

$$c_n = 2s^2 r_{w_0w}^{(2)} [f_w + \partial_{y'}(m-r)r_{w'w''}^{(1)} f_{w''}] d(w-n), \quad (\text{B.11})$$

satisfies the integral equation

$$c_n = l(m-r)r_{w_0n}^{(1)} + \partial_{y_n}(m-r)r_{nw}^{(1)} c_w, \quad (\text{B.12})$$

which can be iterated. Performing Fourier transform in y one finds the initial condition

$$\tilde{c}_n^{(0)} = l(m-r) \exp\left(-\frac{s^2 k^2 t_n}{2} + i h k t_n - r t_n\right). \quad (\text{B.13})$$

and Fourier-transformed Eq(B.12)

$$\tilde{c}_n = \tilde{c}_n^{(0)} + (m-r)(-ik) \int_0^{t_n} dt \exp\left(-\frac{s^2 k^2 (t_n - t)}{2} + i h k (t_n - t) - r(t_n - t)\right) \tilde{c}_w. \quad (\text{B.14})$$

Substituting (B.13) in (B.14) and integrating one finds that each subsequent iteration picks up an additional term $(m-r)(-ik)(t_n^n / n!) \tilde{c}_n^{(0)}$. Collecting the series for the exponent one obtains

$$\tilde{c}_n = l(m-r) \exp\left(-\frac{s^2 k^2 t_n}{2} + i h_r k t_n - r t_n\right), \quad (\text{B.15})$$

where $h_r = r - s^2 / 2$, or

$$c_n = \frac{la}{\sqrt{2\pi}t_n} \exp\left[-\frac{(y_n - h_r t_n)^2}{2s^2 t_n} - r t_n\right], \quad (\text{B.16})$$

and we made use of notation (3.5) $a = (m-r)/s$. Note, that r.h.s. of (B.16) is proportional the Green function of risk-neutral measure. Substituting this solution into Eq(B.11) and dividing both sides on $r_{w_0 w}^{(2)}$ yields the integral equation

$$f_w = \frac{la}{2s} \exp\left[-\frac{a}{s} y + \frac{m^2 - r^2 - s^2(m+3r)}{2s^2} t\right] - a s \partial_y r_{w w}^{(1)} f_w, \quad (\text{B.17})$$

to be solved by the same method of resolvent. Iterating the zeroth approximation given by the inhomogeneous term in (B.17) one finds that each iteration adds up a term $a^{2n}[(T-t)^n / n!] f_w^{(0)}$. Collecting the exponent one obtains the solution

$$f_w = \frac{la}{2s} \exp\left[-\frac{a}{s} y + \frac{m^2 - r^2 - s^2(m+3r)}{2s^2} t + a^2(T-t)\right]. \quad (\text{B.18})$$

Now, using (B.6) gives

$$\bar{F}_w = \frac{l}{2} \exp\left[-\frac{a}{s} y + \frac{m^2 - r^2 - s^2(m+3r)}{2s^2} t\right] \left[e^{a^2(T-t)} - 1\right], \quad (\text{B.19})$$

and (B.7) allows one to find the Lagrange multiplier

$$l = 2F_0 \left[e^{a^2(T-t)} - 1\right]^{-1} \quad (\text{B.20})$$

Substituting this into (B.18) yields Eqs(3.4)-(3.5) in the main text. Finally, (B.19) yields the expected P&L from t to T ,

$$\bar{F}_w = F_0 \exp\left[-\frac{a}{s} y + \frac{m^2 - r^2 - s^2(m+3r)}{2s^2} t\right] \frac{e^{a^2(T-t)} - 1}{e^{a^2 T} - 1}. \quad (\text{B.21})$$

Appendix C. Log-normal Distribution of P&L for the Black-Jones-Perold Portfolio Insurance Strategy

Introducing the “cushion” variable, $x = X - X_0 e^{rt}$, one can rewrite (4.2) in the form

$$\frac{\partial P}{\partial t} + r \frac{\partial(FP)}{\partial F} + m_m x \frac{\partial P}{\partial x} + \frac{1}{2} s_m^2 x^2 \frac{\partial^2 P}{\partial x^2} = 0, \quad (C.1)$$

where $m_m = r + m(m-r)$, $s_m = m s$. This backward log-diffusion equation satisfies the final condition $P(F | x, T) = d(F - x - X_0 e^{rT})$. Introducing a new function

$$P(F | x, t) = e^{r(T-t)} f(z, y, t), \quad (C.2)$$

$$t = T - t, \quad z = \ln \frac{F}{X_0} + r t, \quad y = \ln \frac{x}{X_0} + h_m t, \quad h_m = m_m - \frac{s_m^2}{2}, \quad (C.3)$$

one finds that f satisfies the forward diffusion equation with an initial condition

$$\frac{\partial f}{\partial t} = \frac{s_m^2}{2} \frac{\partial^2 f}{\partial y^2}, \quad f(z, y, 0) = \frac{d[y - \ln(e^z - e^{rT})]}{X_0(e^z - e^{rT})}. \quad (C.4)$$

The solution reads

$$f(z, y, t) = \frac{1}{X_0(e^z - e^{rT})\sqrt{2ps_m^2 t}} \exp\left\{-\frac{[y - \ln(e^z - e^{rT})]^2}{2s_m^2 t}\right\}. \quad (C.5)$$

Using (C.3) and returning to the capital variable X one finds Eq(4.3) in the main text.

Appendix D. Numerical Example of One-Step Hedging.

Let us assume that after performing the Monte Carlo simulation a risk management facility is concerned with two future states of the world. Tomorrow the leveraged model portfolio may have either normal distribution with expected value \$200M and standard deviation \$25M (if the macroeconomic index is unchanged with probability 95%) or the expected value is going to be \$100 with standard deviation \$10M (if the macroeconomic index jumps down, with probability 5%). This portfolio (with its market value) is at risk today. In view of this substantial risk one can neglect details associated with one-day discounting ($Z=1$). Then Eq(5.1) gives the expected value $0.05 \cdot 100 + 0.95 \cdot 200 = \$195M$. The expected variance follows from Eq(5.3)

$$V_w = 0.05 \cdot 10^2 + 0.95 \cdot 25^2 + 0.05 \cdot 0.95 \cdot (200 - 100)^2 = 1073.75, \quad \sqrt{V_w} = \$32.76M.$$

This variance can be minimized by shorting one contract on the index, $j_w = -(200 - 100) / (1200 - 100) = -1$. Then from Eq(5.5) one finds

$$V_{w,j=-1} = 0.05 \cdot 10^2 + 0.95 \cdot 25^2 = 598.75, \sqrt{V_{w,f=-1}} = \$24.47\text{M}.$$

The expected value with such Δ – hedge is given by Eq(5.8)

$$\bar{F}_{w,j=-1} = 1200 \frac{200 - 100}{1200 - 1100} + \frac{200 \cdot 1200 - 100 \cdot 1100}{1200 - 1100} = \$200\text{M}.$$

Given the “sumption that the index may jump down, it is beneficial to short it: not only it reduces variability, but also improves returns.

References

-
- ⁱ E.Derman, *Reflections on Fischer*, J. Portf. Mgmt, Special Issue, Dec 1996, also in http://www.gs.com/qs/doc/reflections_on_fischer.html
- ⁱⁱ S. Esipov and I. Vaysburd, *Time Dynamics of Probability Measure and Hedging of Derivatives*. Archive at Los Alamos (1998) <http://xxx.lanl.gov/abs/math.PR/9805014>
- ⁱⁱⁱ S. Esipov and I. Vaysburd, *On the Profit and Loss Distribution of Dynamic Hedging Strategies*, IJTAF, 2(2) 131-152, 1999, see also http://papers.ssrn.com/paper.taf?ABSTRACT_ID=145172
- ^{iv} P. A. Samuelson, *Collected Scientific Papers of Paul. A. Samuelson*, Cambridge, MA, MIT Press, 1972
- ^v M.F.M. Osborne, *Brownian Motion in the Stock Market*, Operations Research, 7, 145-173, 1959
- ^{vi} H. M. Markowitz, *Portfolio Selection*, J. of Finance, VII, n.1, pp.77-91, 1952
- ^{vii} H. M. Markowitz, *Portfolio Selection: Efficient Diversification of Instruments*, New York, J. Wiley & Sons, 1959
- ^{viii} W. F. Sharpe, *A Simplified Model for Portfolio Analysis*, Management Science, 9, pp.277-293, 1963
- ^{ix} W. F. Sharpe, *Portfolio Theory and Capital Markets*, New York, McGraw Hill, 1970
- ^x S.M. Stigler, *The History of Statistics: The Measurement of Uncertainty Before 1900*, Harvard Univ. Press, 1990
- ^{xi} D.N. Dreman, *Contrarian Investment Strategies : The Next Generation : Beat the Market by Going Against the Crowd*, Simon & Schuster, 1998
- ^{xii} F. Black and R. Jones, *Simplifying Portfolio Insurance*, J. Portfolio. Mgmt, Fall, pp.48-51, 1987
- ^{xiii} F. Black and A.F. Perold, *Theory of Constant Proportion Portfolio Insurance*, J. Economic Dynamics and Control, 16, 403-426, 1992
- ^{xiv} H.E. Leland, *Who Should Buy Portfolio Insurance?*, J. Finance, 35(2), pp.581-596, 1980
- ^{xv} M. Rubinstein, *Alternative Paths to Portfolio Insurance*, Financial Analysts Journal, 41(4), pp.42-52, 1985
- ^{xvi} M. Rubinstein, *Derivatives: A PowerPlus Picture Book*, 1998, <http://www.in-the-money.com>
- ^{xvii} "In times of higher market volatility, trading size is reduced. In losing periods, positions are reduced and trade size is cut back. The central objective is always to preserve capital until more favorable price trends reappear." – from The Turtle Techniques, <http://turtletrader.com/it.html>
- ^{xviii} J. Tobin, *Liquidity Preference as Behavior Toward Risk*, Rev. of Economic Studies, 67, pp.65-86, 1958
- ^{xix} H. Föllmer and P. Leukert, *Quantile Hedging*, preprint 1997
- ^{xx} I. Karatzas and J. Cvitanic, *On Dynamic Measures of Risk*, to be published in Finance and Stochastics, http://papers.ssrn.com/paper.taf?ABSTRACT_ID=89489
- ^{xxi} Private communication. C. Hipp, *Hedging General Claims in Diffusion Models*, U. Karlsruhe preprint 1998; C. Hipp, M. Taksar, *Hedging in Incomplete Markets and Optimal Control*, U. Karlsruhe preprint 1999.